

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	
Project Name	Human Development Index (HDI) Predictor – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper Python indentation followed in all files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into <code>app.py</code> , <code>train_model.py</code> , <code>models/</code> , <code>dataset/</code> , and supporting files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as <code>life_expectancy</code> , <code>gni_per_capita</code> , <code>model</code> , and <code>prediction</code> clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named according to their functionality, improving readability.
5	Code Comments	Inline and block comments explain log	Yes	Important sections of the code are documented with comments.

6	Modular Design	Code is split into reusable functions/m	Yes	Model training and prediction logic are maintained separately for reusability.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code is optimized without unnecessary duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling improve application stability.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Streamlit User Interface	Python, Streamlit	User input and prediction display	high

2	Data Preprocessing Module	Python, Pandas, NumPy	Dataset cleaning and preprocessing	High
3	Machine Learning Model	Scikit-learn	HDI prediction engine	High
4	Model Loader (hdi_model.pkl)	Pickle	Loading trained model during prediction	High
5	Label Encoder (label_encoder.pkl)	Scikit-learn	Decoding predicted HDI category	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project files are well organized with a clear directory structure.
Readability	5	Variable names, function names, and comments make the code easy to understand.
Reusability	5	Prediction, preprocessing, and model loading modules can be reused in future applications.
Documentation / Comments	4	The code contains useful comments explaining major sections and logic.
Overall Score	4.8	The HDI Prediction project follows good coding practices and is maintainable, readable, and reusable.